MILITARY SPECIFICATION

TEST HARNESS AN/AWM-11(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be used to hold, position and interconnect the units of the Fuze Function Control Set AN/AWW-1 for bench testing and alignment.

1.2 Classification - The Test Harness AN/AWM-11 shall consist of the following items:

Item	Type Designation	Applicable Paragraph
Rack, Electrical Equipment	MT-2275/AWM-11	3.4.1
Rack, Electrical Equipment	MT-2277/AWM-11	3.4.2
Distribution Box	J-1131/AWM-11	3.4.3
Cable	W1 of AN/AWM-11	3.4.4.3
Cable	W2 of AN/AWM-11	3.4.4.4
Cable	W3 of AN/AWM-11	3.4.4.5
Cable	W4 of AN/AWM-11	3.4.4.6

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-H-15362	Harnesses, Test Bench, Avionic Equipment, General Specification for
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, Format for

FSC-4940

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MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for
SAR-300	Material Changes and Material Bulletins, Preparation of

STANDARDS

Military

MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC or 115/200 Volt 3-Phase AC, or Combinations

DRAWINGS

60A16C13	W-1 of AN/AWM-11
60A16C14	W-2 of AN/AWM-11
60A16F15	W-3 of AN/AWM-11
60A16C16	W-4 of AN/AWM-11
60A16D9	Distribution Box J-1131/AWM-11
60A16D8	Rack, Electrical Equipment MT-2277/AWM-11
60A16F4	Rack, Electrical Equipment MT-2275/AWM-11

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(2) Copies of applicable drawings may be obtained upon application to the Commanding Officer, U. S. Naval Avionics Facility, Indianapolis 18, Indiana. Requests should be accompanied by justification of need.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200 shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

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(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable subassemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-H-15362 for design, construction, and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment shall be a minimum consistent with good design and shall not exceed 12 pounds.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors conforming with Specification MIL-H-15362.

3.2.3.2 Input Power Connections - Input power connections shall be in accordance with MS 25245.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.5 Maintenance Provisions - Provisions for maintenance shall be as specified in Specification MIL-T-21200.

3.2.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. See 6.4.

3.2.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 VAC and/or 27.5 ± 0.5 VDC

3.2.8 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-H-15362.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

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3.4 Detail Requirements -

3.4.1 Rack, Electrical Equipment MT-2275/AWM-11 - The mounting rack shall meet the following requirements:

3.4.1.1 Function - The Rack, Electrical Equipment MT-2275/AWM-11 shall secure and position the Amplifier, Power Supply AM-2014/AWW-1 and the Oscillator RFO-562/AWW-1 and shall conform to Drawing 60A16F4.

3.4.1.2 Form Factor - This unit is an "L" shaped bracket 9-1/2 inches wide, by 10-1/2 inches long by 5-1/4 inches high.

3.4.1.3 Position - The rack shall position the Amplifier, Power Supply AM-2014/AWW-1 and the Oscillator RFO-562/AWW-1 so as to facilitate the adjustment, alignment, testing under power, repair, and replacement of components.

3.4.2 Rack, Electrical Equipment MT-2277/AWM-11 - The mounting rack shall meet the following requirements:

3.4.2.1 Function - The Rack, Electrical Equipment MT-2277/AWM-11 shall secure and position the Control, Fuze Function C-2612/AWW-1 and shall conform to Drawing 60A16D8.

3.4.2.2 Form Factor - The unit is an inverted "L" shaped bracket 6-1/4 inches wide, by 4-1/4 inches long by 4 inches high.

3.4.2.3 Position - This rack shall position the Control, Fuze Function C-2612/AWW-1 so as to facilitate the adjustment, alignment, testing under power, repair, and replacement of components.

3.4.3 Distribution Box J-1131/AWM-11 - The Distribution Box J-1131/AWM-11 shall meet the following requirements:

3.4.3.1 Function - The Distribution Box J-1131/AWM-11 shall provide for electrically and mechanically joining cable assemblies of the test bench harness and shall conform to Drawing 60A16D9.

3.4.3.2 Form Factor - This unit is rectangular in shape and is 8 inches wide by 7 inches high by 4 inches deep.

3.4.3.3 Controls - Two control switches are located on the Distribution Box J-1131/AWM-11.

3.4.3.3.1 POWER ON-OFF - The "POWER ON-OFF" switch is a single throw toggle switch located in the lower left of the Distribution Box J-1131/AWM-11 and is used to apply both a-c and d-c power to the Fuze Function Control Set AN/AWW-1.

3.4.3.3.2 RELEASE PULSE ON-OFF-ON - The "RELEASE PULSE ON-OFF-ON" switch is a three-position toggle switch located in the lower right of the Distribution Box J-1131/AWM-11 and is used to trigger the output of the Fuze Function Control Set AN/AWW-1. The center position of the switch is the "OFF" position. In the up position the switch is in the "ON" position, while the down position is spring loaded so as to provide contact only as long as the switch is held in the down position.

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3.4.3.4 Electrical Connections - Connections to external circuits shall be as follows:

Reference Designation	Receptacle Type	Function
J101	MS 3102R-14S-5P	115 V INPUT
J102	MS 3102R-10SL-4P	28 V INPUT
J103	MS 3102R-22-14S	POWER OUT

3.4.4 Cable Assemblies -

3.4.4.1 Types - The cable assemblies shall consist of the interconnecting cables W1, W2, W3, and W4 of AN/AWM-11 and shall be as specified in the applicable drawings listed herein.

3.4.4.2 Function - The cable assemblies shall provide for electrical connection of the components of the Fuze Function Control Set AN/AWW-1 to allow normal operation of the set.

3.4.4.3 W1 of AN/AWM-11 - This cable is used to connect the distribution box to the 28 VDC on the test bench and shall conform to Drawing 60A16C13.

3.4.4.4 W2 of AN/AWM-11 - This cable is used to connect the distribution box to the 115 VAC on the test bench and shall conform to Drawing 60A16C14.

3.4.4.5 W3 of AN/AWM-11 - This cable is a multibranch cable used to interconnect the distribution box, the amplifier power supply, and the control box and shall conform to Drawing 60A16F15. It also has a branch to allow power for the Bullpup missile to be connected through the Amplifier, Power Supply AM-2014/AWW-1.

3.4.4.6 W4 of AN/AWM-11 - This cable provides a means of connecting power to the Bullpup missile through the Control, Fuze Function C-2612/AWW-1 and shall conform to Drawing 60A16C16.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

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4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contract shall submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and resubmitted as a production equipment.

4.3 Production Sample Test - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipments for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be interconnected with the components of the Fuze Function Control Set AN/AWW-1 and each operating function shall be checked.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein without failure. A failure shall be defined as anything which causes malfunctioning of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) with a minimum amplitude of 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes. The equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. The direction of vibration shall be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. Should a failure occur, it should be repaired and the test started over again. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. After completion of the manufacturing reliability test, the operation test of paragraph 4.4.1.2 shall be repeated.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the government inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on the results of the preproduction and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 30 hours.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipment previously subjected to sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Test Procedures - The test procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect the weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Use - The test bench harness described by this specification shall be used to interconnect the units of the Fuze Function Control Set AN/AWW-1 and in conjunction with the Fuze Function Control Test Set AN/AWM-4 to bench test and align the Fuze Function Control Set AN/AWW-1.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable specifications. Any deviation from this specification or subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specification - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 The parenthesis (*) when used in the type designation shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.6. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-T-21200 it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

NOTICE - When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government hereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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